Heart disease prevalence

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Instructions

The exam duration is **one hour and a half**.

For this exam, **all documents are allowed** (including all previous notebooks), fact sheets/information sheets you may have prepared.

Your answers must be complete but there is no need to give the code you typed unless otherwise specified. Show your work, including formulas and calculations where appropriate. Interpret all results in the context of Alzheimer's disease research and potential clinical implications.

Note about α threshold: For all statistical tests, use a **significance level of $\alpha = 0.03$**.

Dataset

You are analyzing data from a heart health study conducted on 150 adults aged 25-65. The dataset `heart_health.csv` contains the following variables:

- `patient_id`: Unique patient identifier
- `age`: Age in years
- `gender`: Gender (Male/Female)
- `bmi`: Body Mass Index (kg/m^2)
- `systolic_bp`: Systolic blood pressure (mmHg)
- `diastolic_bp`: Diastolic blood pressure (mmHg)
- `cholesterol`: Total cholesterol level (mg/dL)
- `exercise_freq`: Exercise frequency per week (0-7 days)
- `smoking_status`: Smoking status (Never/Former/Current)
- `heart_disease`: Presence of heart disease (Yes/No)

Questions

Load the dataset using: `heart_data <- read.csv("heart_health.csv")`

```
heart_data = read.csv("heart_health.csv")
```

Describing the dataset

Question 1. (1 point) **No need to give the R code**
How many patients are below 35 years old? **1 point for correct value**

```
nrow(subset(heart_data, age < 35))
```

```
## [1] 34
```

Question 2. (2 points) **No need to give the R code**
What is the average BMI of individuals aged over 60? **2 points for correct value**

```
mean(heart_data$bmi[heart_data$age > 60])
```

```
## [1] 24.96667
```

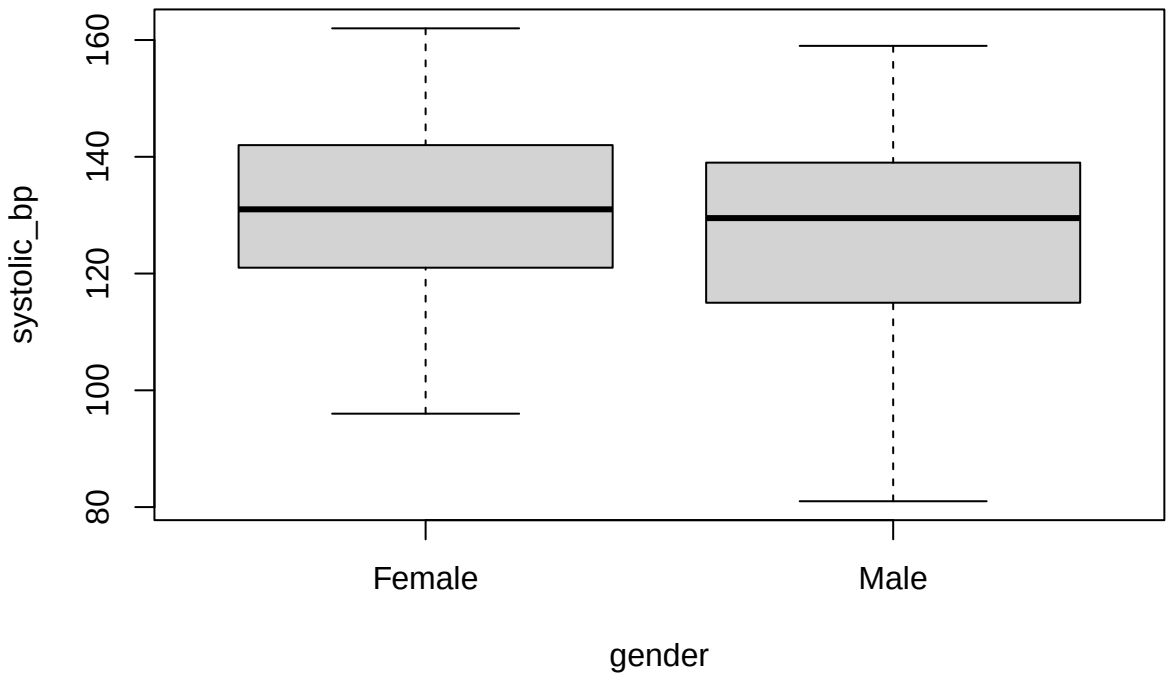
```
# or

mean(heart_data$bmi[heart_data$age >= 60])
```

```
## [1] 25.33333
```

Question 3. (3 points) **Give the R code**
Create a boxplot showing the distribution of systolic blood pressure by gender. Include appropriate labels and title, and interpret the plot in 2-3 sentences discussing any visible differences between groups. **1 point for code, 2 points for interpretation**

```
boxplot(systolic_bp ~ gender, data = heart_data)
```


systolic_bp
80
100
120
140
160
Female
Male
gender

Testing simple hypotheses

Question 4. (3 points)
Test whether there is a difference in mean BMI between patients with and without heart disease. State your hypotheses, perform the test, **give the code you used to perform the test**, and interpret your findings at = 0.03. **1 point for hypotheses, 1 point for the test, 1 point for interpretation**

We want to perform a t-test with two independent samples between patients with and without heart diseases.

Hypotheses: H0 = no difference in BMI between patients H1 = there is a difference in BMI between patients

Check assumptions:

```
table(heart_data$heart_disease)
```

```
##
##  No Yes
## 132  18
```

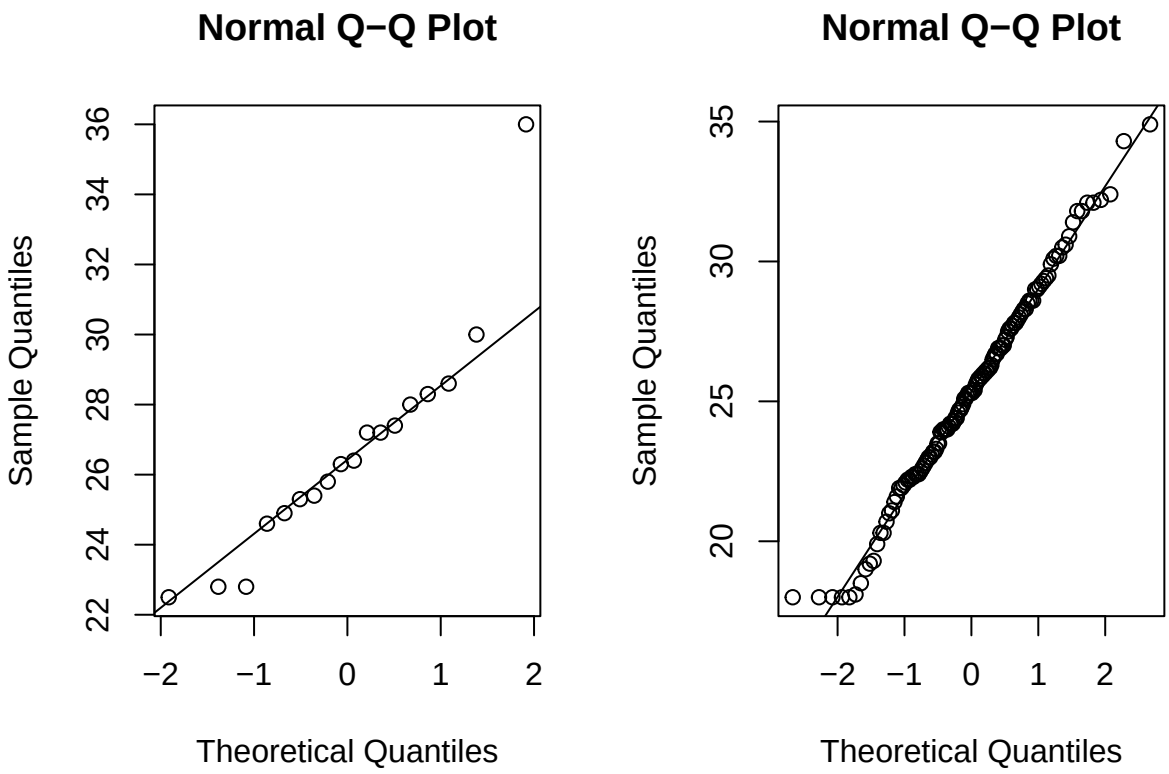
There are 18 patients with heart disease, which is less than 30, as such we need to check the normality of both samples as well as there homoscedasticity.

```
with_disease = subset(heart_data, heart_disease == "Yes")

without_disease = subset(heart_data, heart_disease == "No")

par(mfrow = c(1, 2))
qqnorm(with_disease$bmi)
qqline(with_disease$bmi)

qqnorm(without_disease$bmi)
qqline(without_disease$bmi)
```



Graphically we see that the quantile lines matched -> so we can say that both samples follow a normal distribution. Let's check their homoscedasticity with a Fisher variance test (Hypotheses: H0 = Equal variances, H1 = Different variances):

```
var.test(with_disease$bmi, without_disease$bmi)
```

```
##
##  F test to compare two variances
##
## data:  with_disease$bmi and without_disease$bmi
## F = 0.72599, num df = 17, denom df = 131, p-value = 0.4566
## alternative hypothesis: true ratio of variances is not equal to 1
## 95 percent confidence interval:
##  0.386107 1.676847
## sample estimates:
## ratio of variances
##          0.7259865
```

p »> 0.03, as such we can't reject H0, and the variances of both samples are equal.

In total we can perform a t-test

```
t.test(with_disease$bmi, without_disease$bmi)
```

```
##
##  Welch Two Sample t-test
##
## data:  with_disease$bmi and without_disease$bmi
## t = 1.5595, df = 23.877, p-value = 0.132
```

```
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
##  -0.4060773  2.9141581
## sample estimates:
## mean of x mean of y
##  26.63889  25.38485
```

We get p = 0.13 »> 0.03, thus we can't reject H0 so the means of both samples are equal, which means that there is no difference in BMI between patients with and without heart disease.

Question 5. (4 points)
Investigate whether there is an association between smoking status and heart disease. Perform the appropriate test, give its hypotheses, and interpret your results. What can you conclude about the relationship between smoking and heart disease? **1 point for correct test, 1 point for hypotheses, 1 point interpretation, 1 point for general conclusion**

We want to perform a chi-square test between two qualitative variables (smoking and heart disease), for this all of the expected counts should be above 5.

Let's check it.

```
chisq.test(heart_data$smoking_status, heart_data$heart_disease)$expected
```

```
## Warning in chisq.test(heart_data$smoking_status, heart_data$heart_disease):
## L'approximation du Chi-2 est peut-être incorrecte
```

```
##                          heart_data$heart_disease
## heart_data$smoking_status    No  Yes
##                   Current 24.64 3.36
##                   Former  45.76 6.24
##                   Never   61.60 8.40
```

We see that there are less than 5 expected people which are current smokers with heart disease. Thus we're currently violating the assumptions of the chi-square test.

To avoid that, we need to regroup Current and Former smokers.

```
heart_data$new_smoking = ifelse(heart_data$smoking_status == "Current", "Former", heart_data$smoking_sta

chisq.test(heart_data$new_smoking, heart_data$heart_disease)$expected
```

```
##                       heart_data$heart_disease
## heart_data$new_smoking   No Yes
##                 Former 70.4 9.6
##                 Never  61.6 8.4
```

Let's now perform the chi-square independence test.

Hypotheses : H0 = heart disease are independent of smoking, H1 = heart disease and smoking are linked

```
chisq.test(heart_data$new_smoking, heart_data$heart_disease)
```

```
##
##  Pearson's Chi-squared test with Yates' continuity correction
##
## data:  heart_data$new_smoking and heart_data$heart_disease
## X-squared = 1.1186, df = 1, p-value = 0.2902
```

We observe p = 0.29 »> 0.03, so we can't reject H0, and as such we don't observe a link between smoking status and heart disease in our sample. This seems contradictory to common knowledge.

Modeling cholesterol levels

Question 6. (7 points)
Build a linear regression model to predict cholesterol levels using age and BMI as predictors.

a) **Give the R code you use to fit the model** (1 point) ** 1 point correct code **

```
lm_chol = lm(cholesterol ~ age + bmi, data = heart_data)
```

b) Report the regression equation with coefficient values (2 points) **2 points for correction equation and coefficients**

```
summary(lm_chol)
```

```
##
## Call:
## lm(formula = cholesterol ~ age + bmi, data = heart_data)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -87.69 -20.24  -2.08  22.50  96.63
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 176.2187    21.7289   8.110 1.83e-13 ***
## age           0.7654     0.2280   3.356  0.00101 **
## bmi           0.4670     0.7500   0.623  0.53448
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 33.18 on 147 degrees of freedom
## Multiple R-squared:  0.07433,    Adjusted R-squared:  0.06174
## F-statistic: 5.902 on 2 and 147 DF,  p-value: 0.003424
```

$$cholesterol = 176 + 0.77 \times age + 0.47 \times bmi$$

c) Interpret the coefficient for age in the context of the study (2 points) **2 points for correct interpretation**

We have a positive coefficient of age, which means that age increases cholesterol content. With each additional year of age, the concentration of cholesterol increases by 0.77 mg/dL.

d) Test the overall significance of the model and report the R^2 value, what can you say about the model? (2 points) **1 point for reporting correct values of R^2 and significance, 1 point for interpretation of the model**

As given by the model, the adjusted R square is 0.06 (=6%) which is very low. It means that using age and BMI explains only 6% of the variation of cholesterol content. The overall p-value of the variance test is equal to 0.003 which is lower than 0.03, so this means that the model is better at explaining cholesterol content than a flat line.

Bonus Questions

(1 point each, not counted in the main 20-point total)

These questions should be directly answered on the answer sheet!

Bonus Question 1. (1 point)
In a clinical trial comparing two treatments, why might researchers choose a paired t-test instead of a two-sample t-test? **1 point for naming measuring same patients twice**

Bonus Question 2. (1 point)
A researcher finds a correlation coefficient of r = 0.85 between two variables and concludes that one variable causes the other. What is wrong with this conclusion? Name one alternative explanation for this strong correlation. **1 point for causation correlation, spurrious correlation, etc.**